Ex.No:06 Error Dection & Error correction

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AIM:

Write a program to implement error detection and correction using HAMMING code concept. Make a test run to input data stream and error correction features.

Sender:

*. Input to sender file should be a text of any length. Program should convert the text to binary.

* Apply Hamming code concepts on the binary data and add Redundant bits to it.

code:

```
data = input("Enter a data:")
m = len(data)
r=0
while 2**r < (m+r+1):
    r+=1
hamming = [0]* (m+r+1)
j=0
for i in range(1, m+r+1):
    if i & (i-1) != 0:
        hamming[i] = int(data[j])
        j+=1
```

```
for i in range(3):
    pos = 2**i
    parity = 0
    for j in range(1, m+r+1):
        if j & pos != 0
            parity ^= hamming[j]
    hamming[pos] = parity.
```

Receiver:

Receiver program should read the input from channel file. Apply hamming code on the binary data to check for error.

If there is an error, display the position of the Error.

Else remove the redudant bits and convert the binary data to ascii & Display the Output.

Code:

```
.Receiver = input("Enter the received Hamming code")
 n = len(receiver)
 hamming = [0] * (n+1)
 for i in range(1, n+1):
     hamming[i] = int(receiver[n-i])
 r=0
 while 2**r < n+1:
     r+=1
 error = 0.
```

```
for i in range(r):
    pos = 2**i
    parity = 0
    for j in range(1, n+1):
        if j & pos != 0:
            parity ^= hamming[j]
    error += parity * pos
if error != 0:
    print("Error detected at position:", error)
else:
    print("There is No Error")
for i in range(n, 0, -1):
    print(hamming[i], end="")
print()
```

Sample Input:

Data bits: 1011

Hamming code: 1100110

Received Hamming code: 1000110

Error detected at position: 6

Correct Hamming code: 1100110

Result:

Error Detection & Error Correction using Hamming code are learned & execute Successfully.